# Multi-Objective Linear Programming Problem

In many real-life optimization problems, there is often more than one objective function to be optimized simultaneously. Multi-Objective Linear Programming (MOLP) deals with optimizing two or more linear objective functions subject to a common set of linear constraints.

A general form of a two-objective linear programming problem can be written as:

$$\boldsymbol{Max/Min\ Z = \left(Z_1, Z_2, \dots, Z_p\right)}$$

$$\textit{\textbf{Subject to:}} \quad \boldsymbol{Ax \le b \;\; or \;\; Ax \ge b \;\; or \;\; Ax = b, \;\; x \ge 0.}$$

*Where* $\boldsymbol{Z_k = a_k x_1 + b_k x_2, \;\; k = 1, \dots, p.}$

**Unlike a single-objective Linear Programming Problem (LPP), where a single optimal solution exists, an MOLP generally yields a set of *efficient* or *Pareto-optimal solutions.***

**Efficient Solution**: Consider the following multi-objective linear programming problem:

$$Min_S\; Z = \left(Z_1, Z_2, \dots, Z_p\right)$$

Where S is the feasible region subset of $\mathbb{R}^n$. $\mathrm{X}^*$ is called an efficient solution if there does not exist $X \in S$ such that

$$Z(X) \leq Z(X^*) = \left(Z_1(X^*), \dots, Z_p(X^*)\right)$$

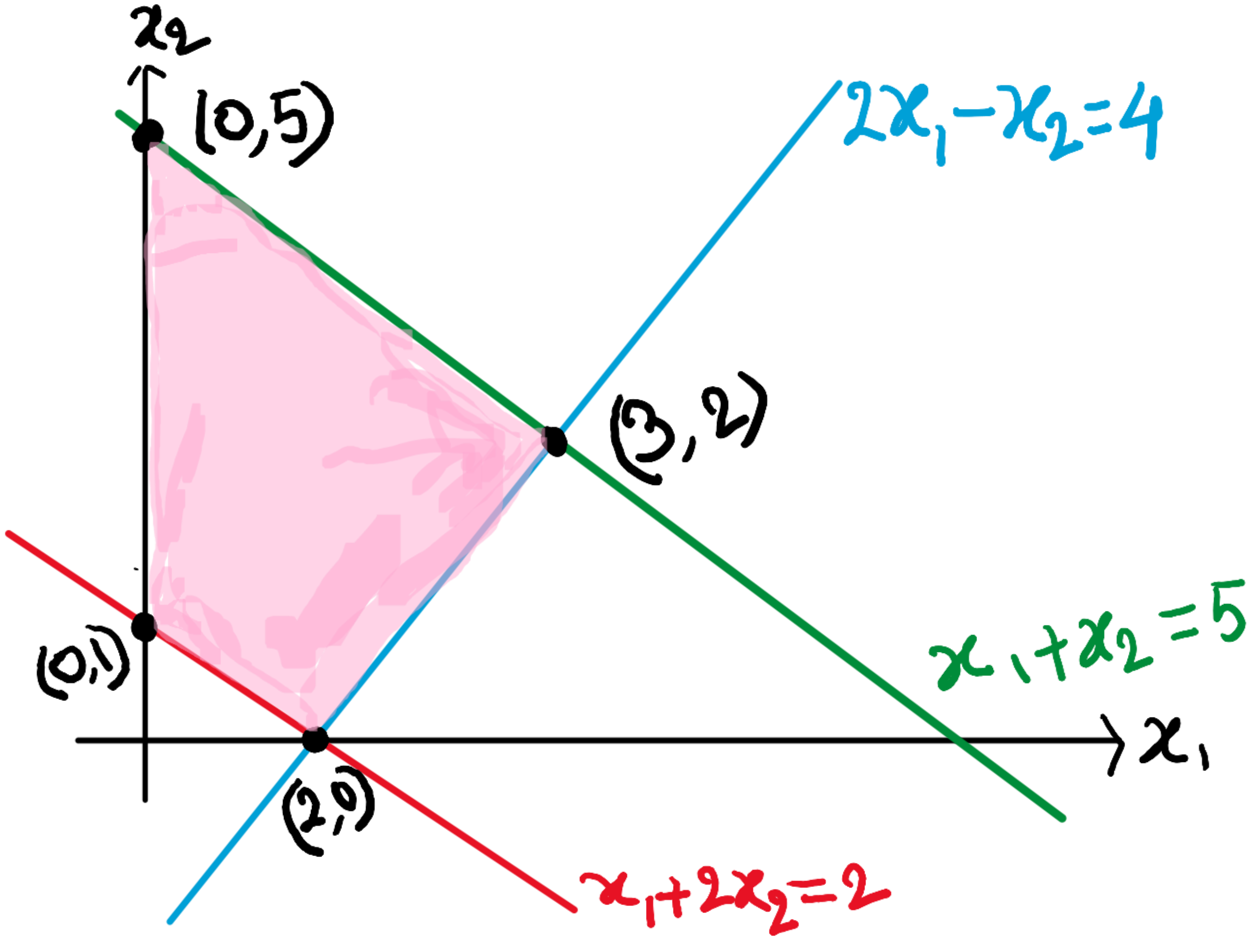
**Efficient Frontier**: The set of all efficient solution of a MOLP is called efficient frontier.

**Ideal Solution:** A solution is an ideal solution if it is feasible and optimal for all objective functions $Z_k$.

$$Min\ Z = \{x_1 + x_2, 5x_1 + 2x_2\}$$

Subject to:

$$x_1 + 2x_2 \geq 2$$
$$2x_1 - x_2 \leq 4$$
$$x_1 + x_2 \leq 5, \quad x_1, x_2 \geq 0$$




| | $Z_1$ | $Z_2$ |
|---|---|---|
| (0,5) | 5 | 10 |
| (0,1) | $\boxed{1}$ | $\boxed{2}$ |
| (2,0) | 2 | 10 |
| (3,2) | 5 | 19 |

**Therefore, $(0, 1)$ be an ideal solution**

$$Min\ Z = \{-2x_1 - x_2, -x_1 - 2x_2\}$$

Dr. Arijit Das

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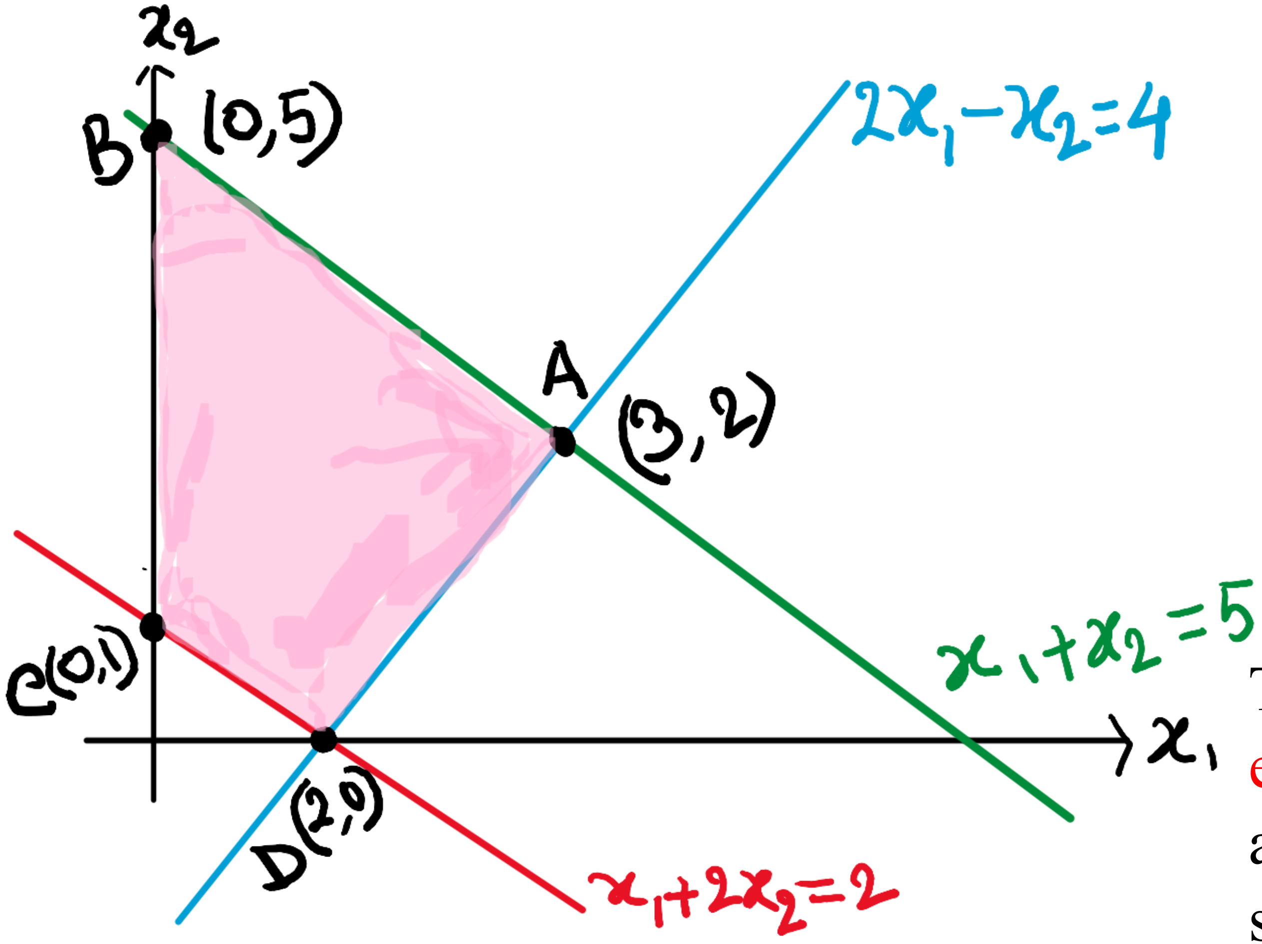
$$2x_1 - x_2 \leq 4$$
$$x_1 + x_2 \leq 5, \quad x_1, x_2 \geq 0$$




| | $Z_1$ | $Z_2$ |
|---|---|---|
| B(0,5) | -5 | $\boxed{-10}$ |
| C(0,1) | -1 | -2 |
| D(2,0) | -4 | -2 |
| A(3,2) | $\boxed{-8}$ | -7 |

The problem have no ideal solution, but two efficient solution B(0, 5) and A(3, 2). Therefore, any point on the line segment AB is an efficient solution. The line is called efficient frontier.